Petar Nizić-Nikolac

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Personal info	Born in 2000, Zagreb, Croatia
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Contact	+41 44 633 8192, petar.nizic-nikolac@ifor.math.ethz.ch
Webpage	https://pnizicnikolac.github.io/
Research interests	High-Dimensional Probability, Statistics, Optimization, Computer Science, Combinatorics

Education

ETH Zürich	PhD in Mathematics, advised by Afonso S. Bandeira	*10/2022 – xy/2027*
Simons Institute for the Theory of Computing	Long-term Visitor at the *Spectral Theory Beyond Graphs* program	*fall/2026*
Trinity College, Cambridge	MMath (Master of Mathematics): Honours with Distinction	*09/2021 – 06/2022*
Trinity College, Cambridge	BA (Bachelor of Arts) in Mathematics: First Class	*09/2018 – 06/2021*
XVth Gymnasium Zagreb	Mathematical grammar school: Student of the Generation award	*09/2014 – 06/2018*

Research

also at Google Scholar

Preprints	Matrix Concentration and Equivalent Operators on Fock Spaces. (with A. S. Bandeira, D. Kunisky, L. Pesenti, and R. Wang). arXiv:2610.01982.
	Convergence of Kikuchi matrices to Γ-independent and q-Gaussian limits. (with A. S. Bandeira, D. Kunisky, L. Pesenti, and R. Wang). arXiv:2610.02008.
	The Hypergraph Moore Bound. (with A. S. Bandeira, D. Kunisky, L. Pesenti, and R. Wang). arXiv:2607.14068.
Publications	Matrix Chaos Inequalities and Chaos of Combinatorial Type. (with A. S. Bandeira, K. Lucca, and R. van Handel). STOC 2025.
	The bunkbed conjecture holds in the $p \uparrow 1$ limit. (with T. Hutchcroft and A. Kent). Combinatorics, Probability and Computing. 2023;32(3):363–369.
Other writing	Randomstrasse101: Open Problems of 2025. (with A. S. Bandeira, D. Dmitriev, K. Lucca, and A. Rödder). arXiv:2603.29571.
	Exercises in Mathematics of Data Science. (with A. S. Bandeira, P. A. Teixeira, K. Lucca, and A. Kireeva). Available online.
In preparation	Structure-aware bounds for random tensor-structured sketching. (with G. Paouris and L. Rebrova). Available soon.
	The Spectral Distribution of Matrix Least Squares Solutions, and the Ellipsoid Fitting Problem. (with A. S. Bandeira and Y. M. Lu). Available soon.

Mentoring

co-advised with Afonso S. Bandeira

Master theses	Shawn Lim (*ongoing*)
	Pavel Kliska (*2025-2026*) → Research Intern at University of Zürich
	Mael Macuglia (*2024*) → Research Intern at University of Zürich
	Xinmeng Zeng (*2023*) → PhD at Stanford University
Bachelor theses	Lukas Münzel (*2026*) → Founding Member at Mirendil
	Stefan Hadzhistoykov (*2024*) → Options Trader at Maverick Derivatives
Semester projects	Suei-Wen Chen (*2025*), Vlad Burca (*2025*), Nicolas Hotton (*2023*), Danielle Ludin (*2023*)

Teaching

Courses	*Spring 2027*: Noncommutative Probability	*co-lecturer*
	Spring 2026: Student Seminar on MDS: Randomized Linear Algebra	*co-advisor*
	Fall 2025: Mathematics of Data Science	*course organizer*
	Spring 2025: Mathematics of Signals, Networks, and Learning (video)	*course organizer*
	Fall 2024: Mathematics of Data Science	*course organizer*
	Spring 2024: Mathematics of Signals, Networks, and Learning	*course organizer*
	Fall 2023: Mathematics of Data Science	*teaching assistant*
	Spring 2023: Mathematics of Signals, Networks, and Learning	*teaching assistant*
Seminar organization	*10/2023 – 08/2026*, DACO Seminar	

Service

Member	VMM (Verein des akademischen Mittelbaus): Executive Board	*03/2026 – ongoing*
	FIM Nachdiplom Committee	*10/2025 – ongoing*
Reviewer	Annals of Probability, Communications of the AMS, Information and Inference: A Journal of the IMA, Symposium on Discrete Algorithms	

Work Experience

G-Research	Quant Research Intern	*London, UK, 06/2025 – 08/2025*
Squarepoint Capital	Quant Software Engineer Intern	*London, UK, 06/2021 – 08/2021*
University of Cambridge	Summer Research Intern (advisor: Tom Hutchcroft)	*Cambridge, UK, 06/2020 – 09/2020*

Outreach

Memberships	Croatian Mathematical Society • Team Selection Test Committee • Deputy Leader at MEMO 2022, Switzerland	*09/2021 – ongoing*
	Society of Young Gifted Mathematicians "Marin Getaldić" • European Mathematical Cup Jury (≈ 700 students) • Organizing camps and popularizing mathematics	*07/2018 – 08/2022*
	Cambridge University Croatian Society • Junior Treasurer	*09/2020 – 06/2022*
Instructor	XVth Gymnasium Zagreb • Preparing students from my former high school for math competitions	*09/2018 – 04/2019*

Honors

PhD grants	G-Research Travel Allowance	*London, UK, 2025*
University awards	Senior Scholar Award	*Cambridge, UK, 2021*
	F. J. Woods Fund Studentship	*Cambridge, UK, 2020*
	CMS Summer Studentship	*Cambridge, UK, 2020*
	Junior Scholar Award	*Cambridge, UK, 2019*
International competitions	Bronze medal at the 59th IMO	*Romania, 2018*
	Bronze medal at the 58th IMO	*Brazil, 2017*
	Team gold and individual gold medals at the 10th MEMO	*Austria, 2016*
	Silver medal at the 12th IJSO	*South Korea, 2015*
	Team gold and individual silver medals at the 9th MEMO	*Slovenia, 2015*

Communications

Talk (inv.)	Seminar in high dimensional probability and statistics, Michigan State University	*12/2026*
Talk (inv.)	Probability Seminar, UC Davis	*10/2026*
Talk (inv., video)	Probabilistic Phenomena on Graphs, Groups, and Manifolds, Simons Institute	*10/2026*
Talk (inv.)	Seminar, Yale	*08/2026*
Talk (inv.)	Seminar, Institute of Science and Technology Austria	*07/2026*
Talk (inv.)	Foundations of Computational Mathematics 2026, Vienna	*07/2026*
Poster	RandNLA workshop, Lausanne	*06/2026*
Talk (inv.)	SIAM Conference on Optimization, Edinburgh	*06/2026*
Talk (inv., video)	1W-MINDS Seminar, online	*10/2025*
Talk	57th Annual ACM Symposium on Theory of Computing, Prague	*06/2025*
Poster	Towards a Theory for Typical-Case Algorithmic Hardness, Les Houches	*02/2025*
Talk (inv.)	Seminar za teoriju vjerojatnosti, PMF Zagreb	*11/2024*
Poster	Mathematical Aspects of Learning Theory – 20 years later, Barcelona	*09/2024*
Talk	Seminar, MIT	*05/2024*
Poster	Citadel Securities PhD Summit, London	*03/2024*
Talk (inv.)	Research Seminar Noncommutative and Functional Analysis, Saarland University	*11/2023*
Talk	Graduate Seminar in Probability, ETH Zürich	*12/2022*
Talk	Summer Research Festival 2020, Cambridge	*10/2020*

Knowledge and Skills

Languages	English (fluent), Croatian (native), German (basic)
Computer skills	C++, Python, MATLAB, Q/kdb+, LaTeX, GitHub